Optimal photosynthetic strategies

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Outline

- Eco-evolutionary optimality theory
- Study 1: Global optimal photosynthetic strategies
- Study 2: Site-level optimal photosynthetic strategies
- Implications

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Eco-evolutionary optimality theory

"EEO invokes the power of natural selection to eliminate uncompetitive trait combinations, and thereby shape predictable, general patterns in vegetation"

New Phytologist

Tansley review | Free Access

Eco-evolutionary optimality as a means to improve vegetation and land-surface models

Sandy P. Harrison, Wolfgang Cramer, Oskar Franklin, Iain Colin Prentice, Han Wang, Åke Brännström, Hugo de Boer, Ulf Dieckmann, Jaideep Joshi, Trevor F. Keenan, Aliénor Lavergne ... See all authors

First published: 15 June 2021 | https://doi.org/10.1111/nph.17558 | VIEW METRICS

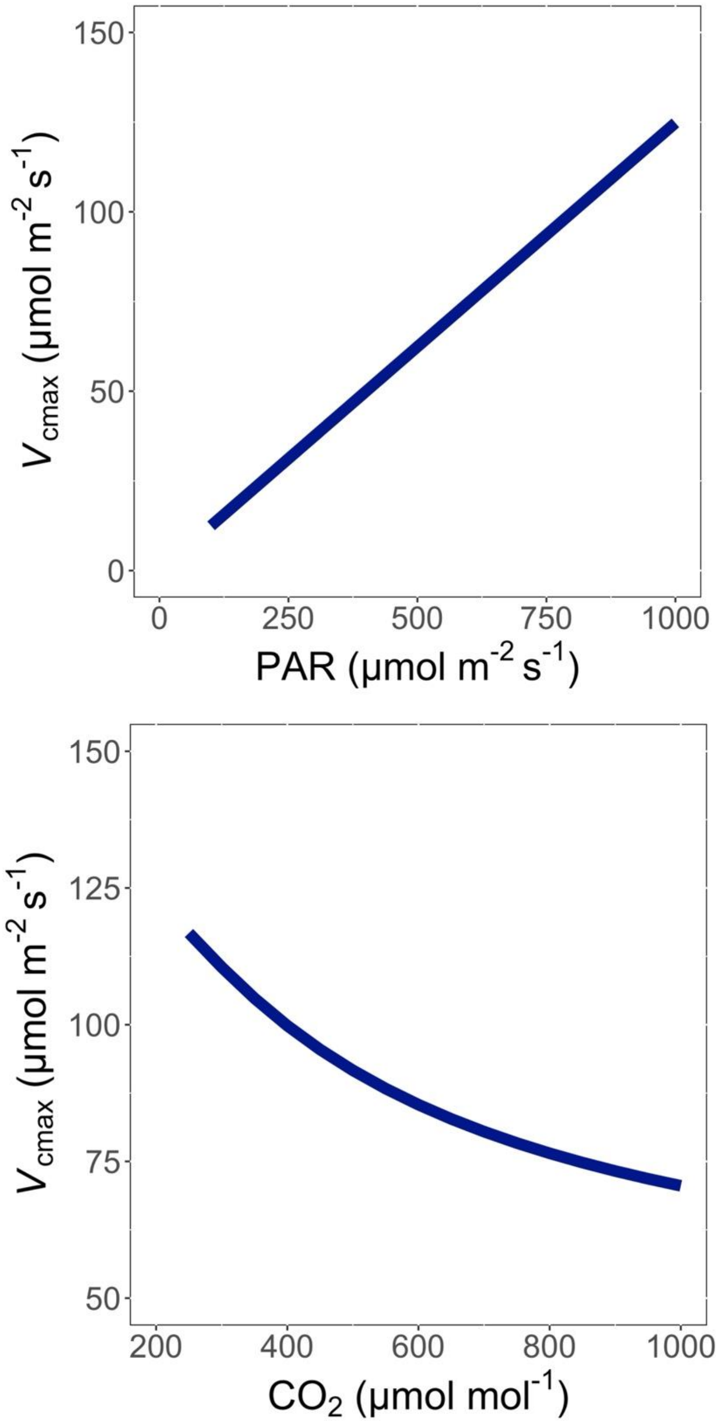
Photosynthesis optimization: least cost

Optimally, plants will maintain fastest rate of photosynthesis at the lowest summed resource cost (water and nutrient use)

Optimal trait example

The maximum rate of RuBisCO carboxylation (V_{cmax})

Smith et al. (2019)

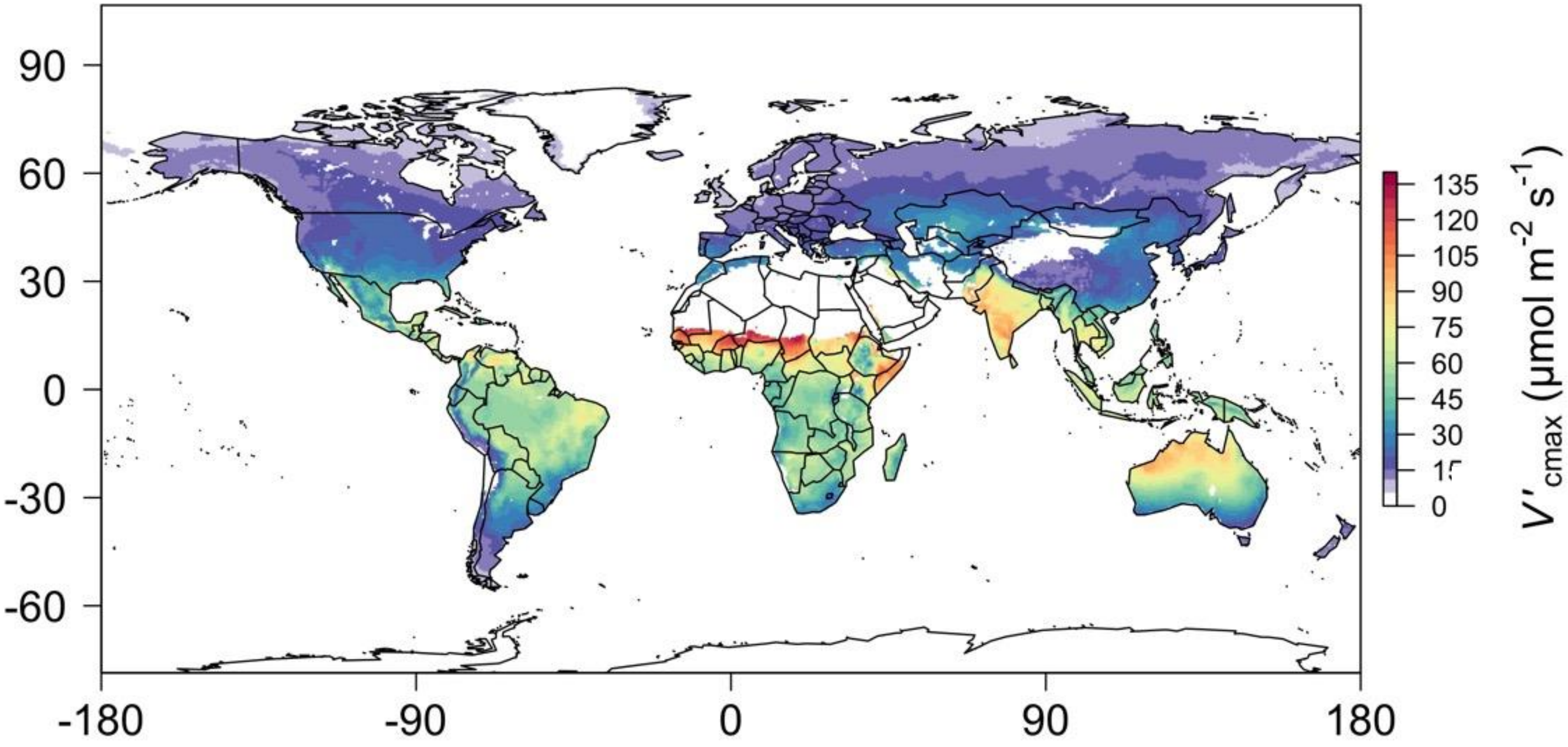


Trait-environment relationships can be predicted in ways that are:

1. Parameter-sparse
2. Mechanistically interpretable

Smith et al. (2019)

We can predict optimal traits in different environments



Smith et al. (2019)

We can predict optimal traits in different environments

Global Change Biology

PRIMARY RESEARCH ARTICLE | Full Access

Mechanisms underlying leaf photosynthetic acclimation to warming and elevated CO_2 as inferred from least-cost optimality theory

Nicholas G. Smith, Trevor F. Keenan

First published: 11 June 2020 | https://doi.org/10.1111/gcb.15212 | VIEW METRICS

JAMES Journal of Advances in Modeling Earth Systems

Research Article | Open Access

A Model of C_4 Photosynthetic Acclimation Based on Least-Cost Optimality Theory Suitable for Earth System Model Incorporation

Helen G. Scott, Nicholas G. Smith

First published: 04 February 2022 | https://doi.org/10.1029/2021MS002470 | VIEW METRICS

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nature > nature plants > letters > article

Letter | Published: 04 September 2017

Towards a universal model for carbon dioxide uptake by plants

Han Wang, I. Colin Prentice, Trevor F. Keenan, Tyler W. Davis, Ian J. Wright, William K. Cornwell, Bradley J. Evans & Changhui Peng

Nature Plants 3, 734–741 (2017) | Cite this article

Science Advances

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HOME > SCIENCE ADVANCES > VOL. 9, NO. 3 > LEAF ECONOMICS FUNDAMENTALS EXPLAINED BY OPTIMALITY PRINCIPLES

RESEARCH ARTICLE | PLANT SCIENCES

Leaf economics fundamentals explained by optimality principles

HAN WANG, I. COLIN PRENTICE, IAN J. WRIGHT, DAVID I. WARTON, SHENGCHAO QIAO, XIANGTAO XU, JIAN ZHOU, KIHACHIRO KIKUZAWA, AND NILS CHR. STENSETH | Authors Info & Affiliations

New Phytologist

Full paper | Open Access

Reduced global plant respiration due to the acclimation of leaf dark respiration coupled with photosynthesis

Yanghang Ren, Han Wang, Sandy P. Harrison, I. Colin Prentice, Owen K. Atkin, Nicholas G. Smith, Giulia Mengoli, Artur Stefanski, Peter B. Reich

First published: 28 October 2023 | https://doi.org/10.1111/nph.19355 | VIEW METRICS

ECOLOGY LETTERS

Letter

Balancing the costs of carbon gain and water transport: testing a new theoretical framework for plant functional ecology

I. Colin Prentice, Ning Dong, Sean M. Gleason, Vincent Maire, Ian J. Wright

First published: 11 November 2013 | https://doi.org/10.1111/ele.12211 | VIEW METRICS

ECOLOGY LETTERS

Letter | Open Access

Global photosynthetic capacity is optimized to the environment

Nicholas G. Smith, Trevor F. Keenan, I. Colin Prentice, Han Wang, Ian J. Wright, Ülo Niinemets, Kristine Y. Crous, Tomas F. Domingues, Rossella Guerrieri, F. Yoko Ishida, Jens Kattge ... See all authors

First published: 04 January 2019 | https://doi.org/10.1111/ele.13210 | VIEW METRICS

How does the environment shape optimal photosynthesis in a multi-trait context?

How does the environment shape optimal photosynthetic strategies?

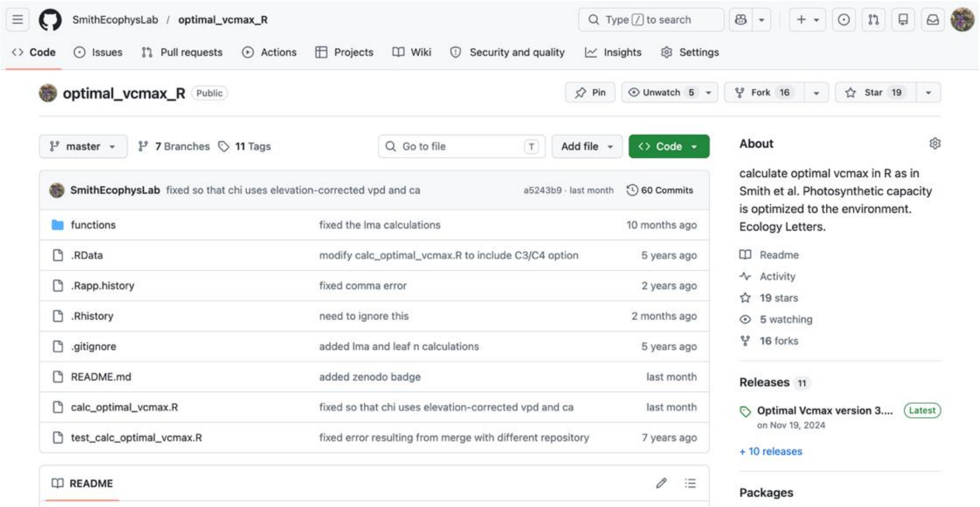
Outline

- Eco-evolutionary optimality theory
- Study 1: Global optimal photosynthetic strategies
- Study 2: Site-level optimal photosynthetic strategies
- Implications

Global simulation strategy

Global simulation strategy

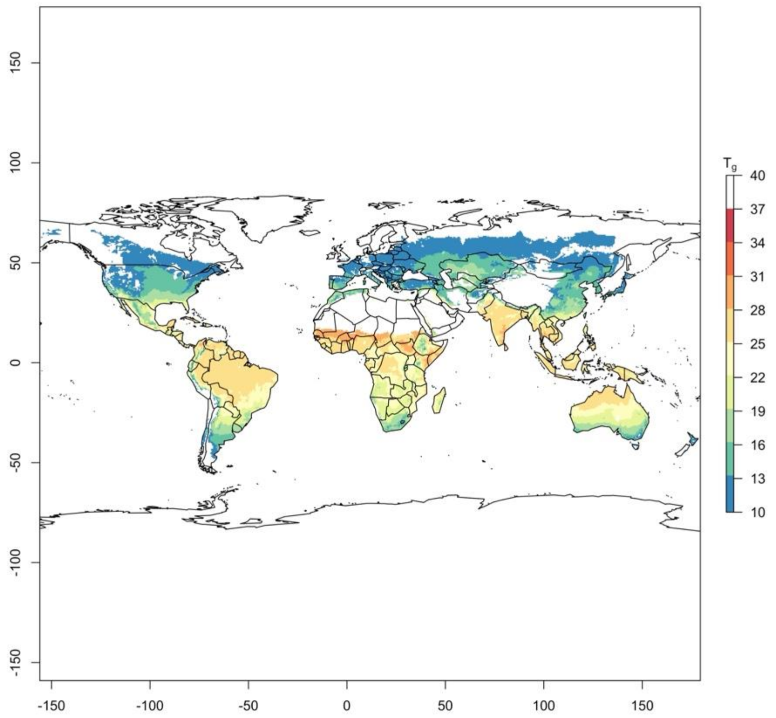
1. Developed optimal photosynthesis model

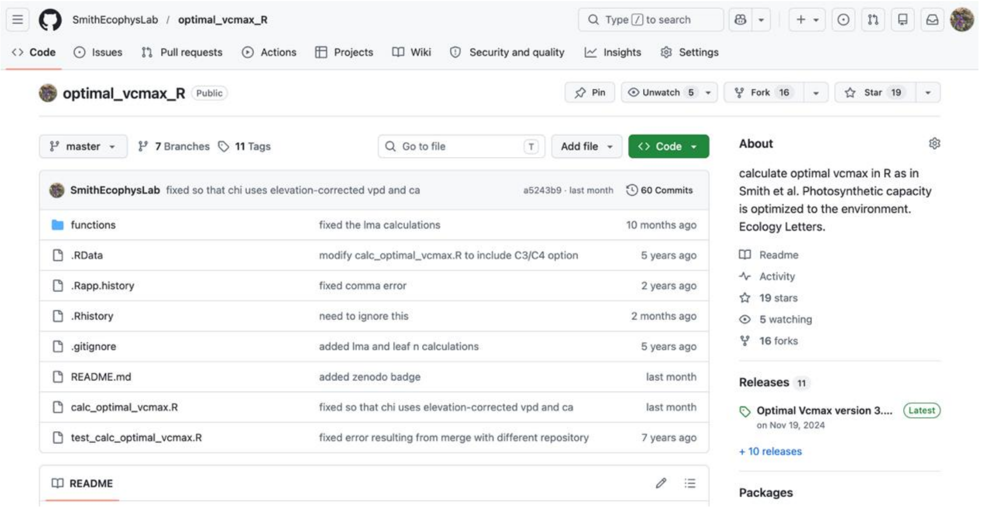


github.com/smithecophyslab/optimal_vcmax_r

Global simulation approach

1. Developed optimal photosynthesis model
2. Used globally gridded climate to simulate traits

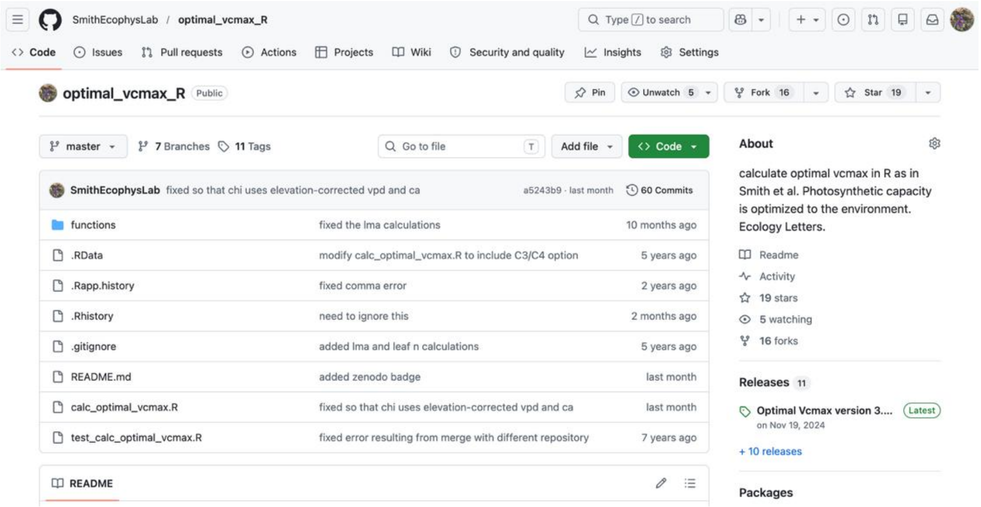




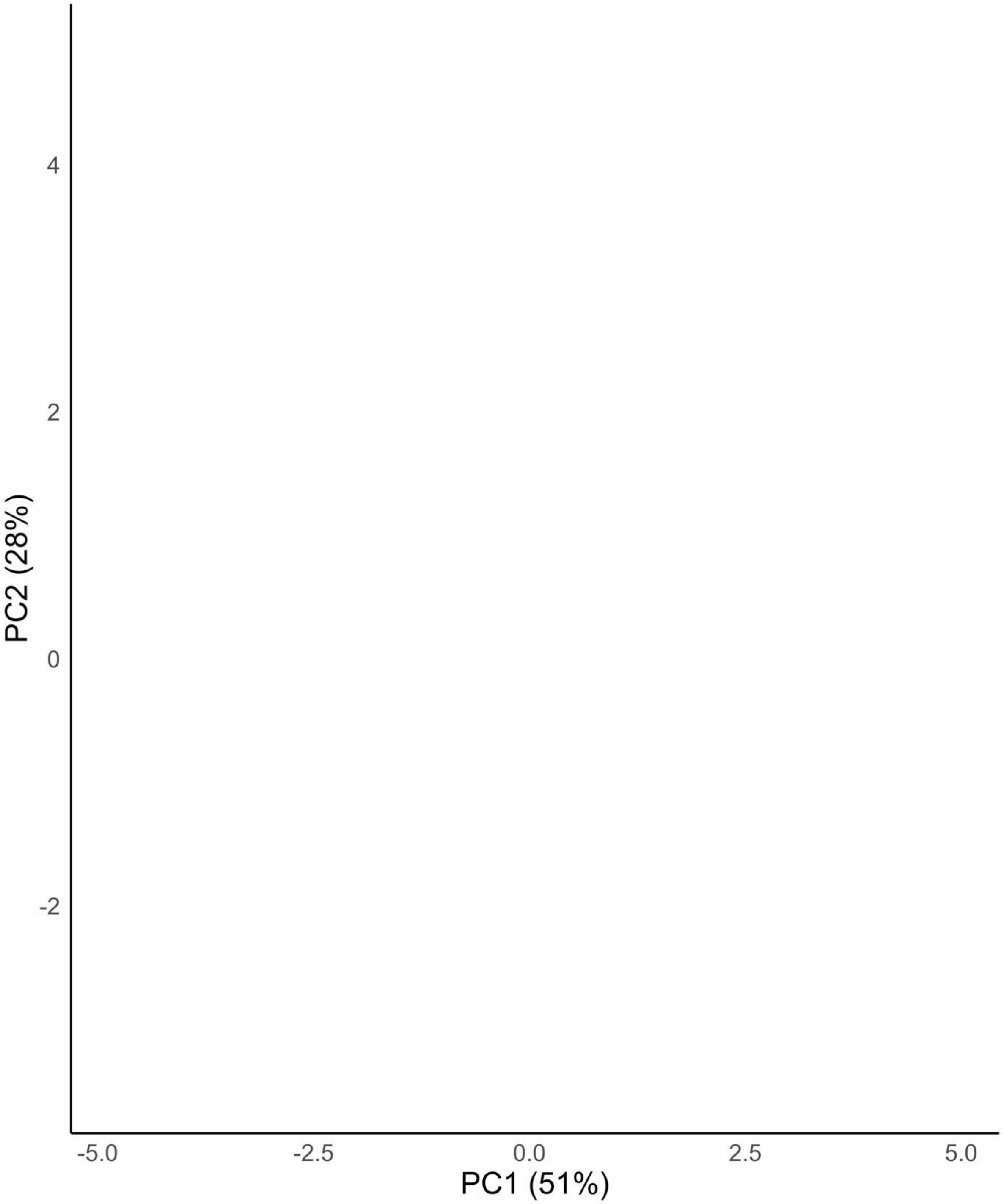
github.com/smithecophyslab/optimal_vcmax_r

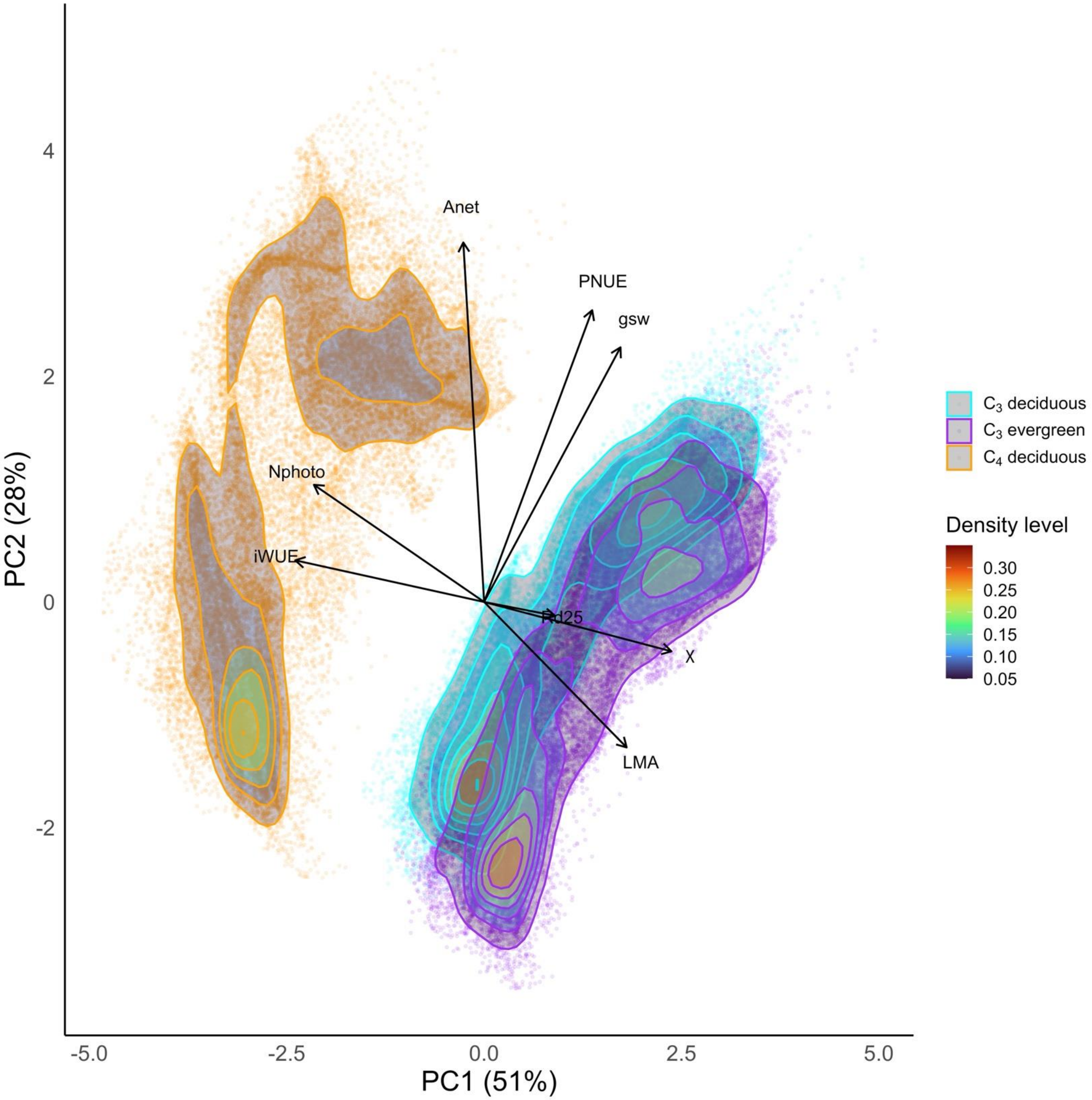
Global simulation approach

1. Developed optimal photosynthesis model
2. Used globally gridded climate to simulate traits
3. Used PCA to explore trait relationships and clustering
    - Carbon uptake (A_{net})
    - Carbon loss (R_d)
    - Water use for photosynthesis (g_{sw})
    - Nitrogen use for photosynthesis (N_{photo})
    - Leaf economics (LMA)
    - Least-cost theory (χ)
    - Photosynthetic efficiency (iWUE, PNUE)

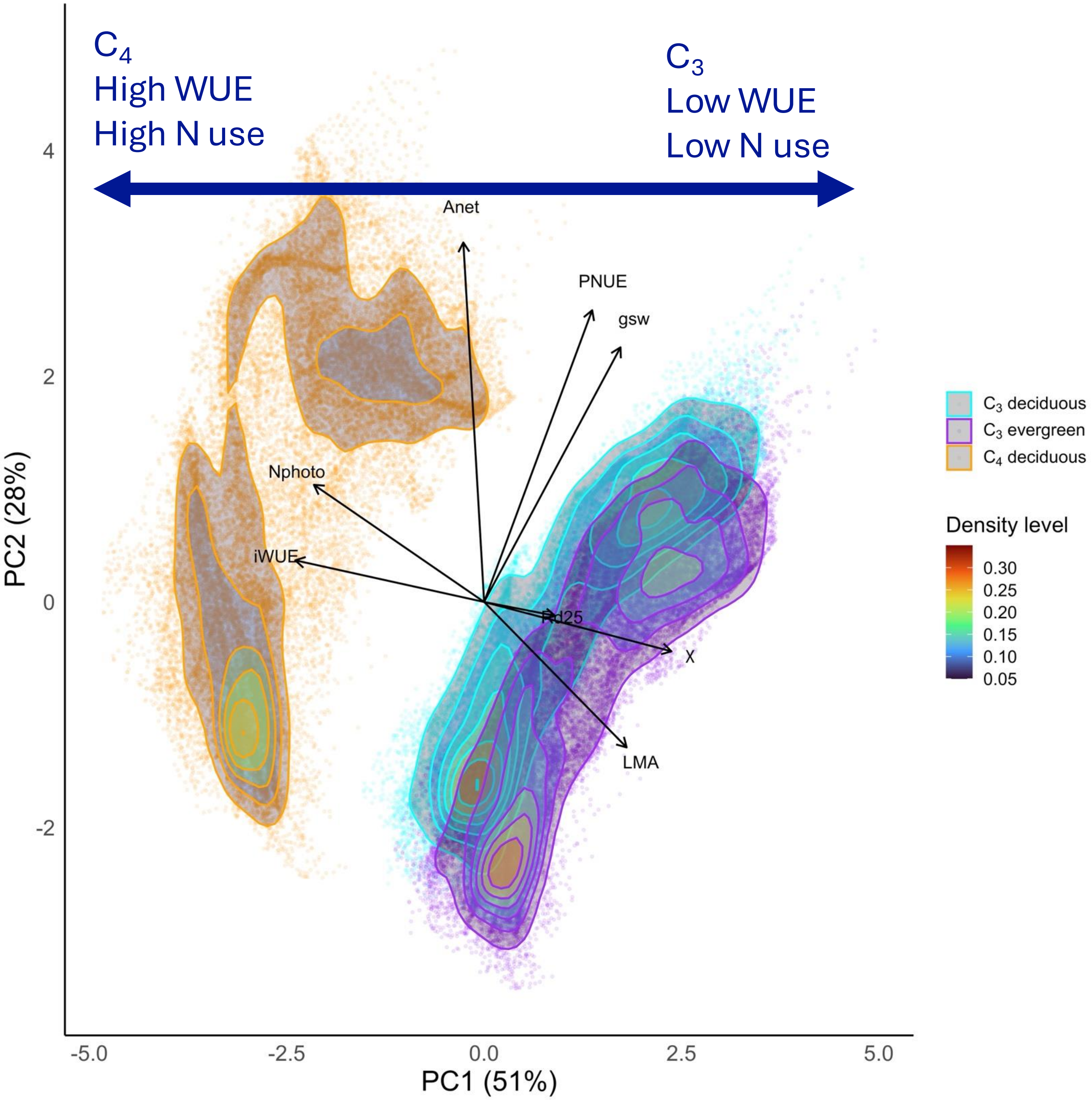


github.com/smithecophyslab/optimal_vcmax_r


PC2 (28%)
4
2
0
-2
-5.0
-2.5
0.0
2.5
5.0
PC1 (51%)

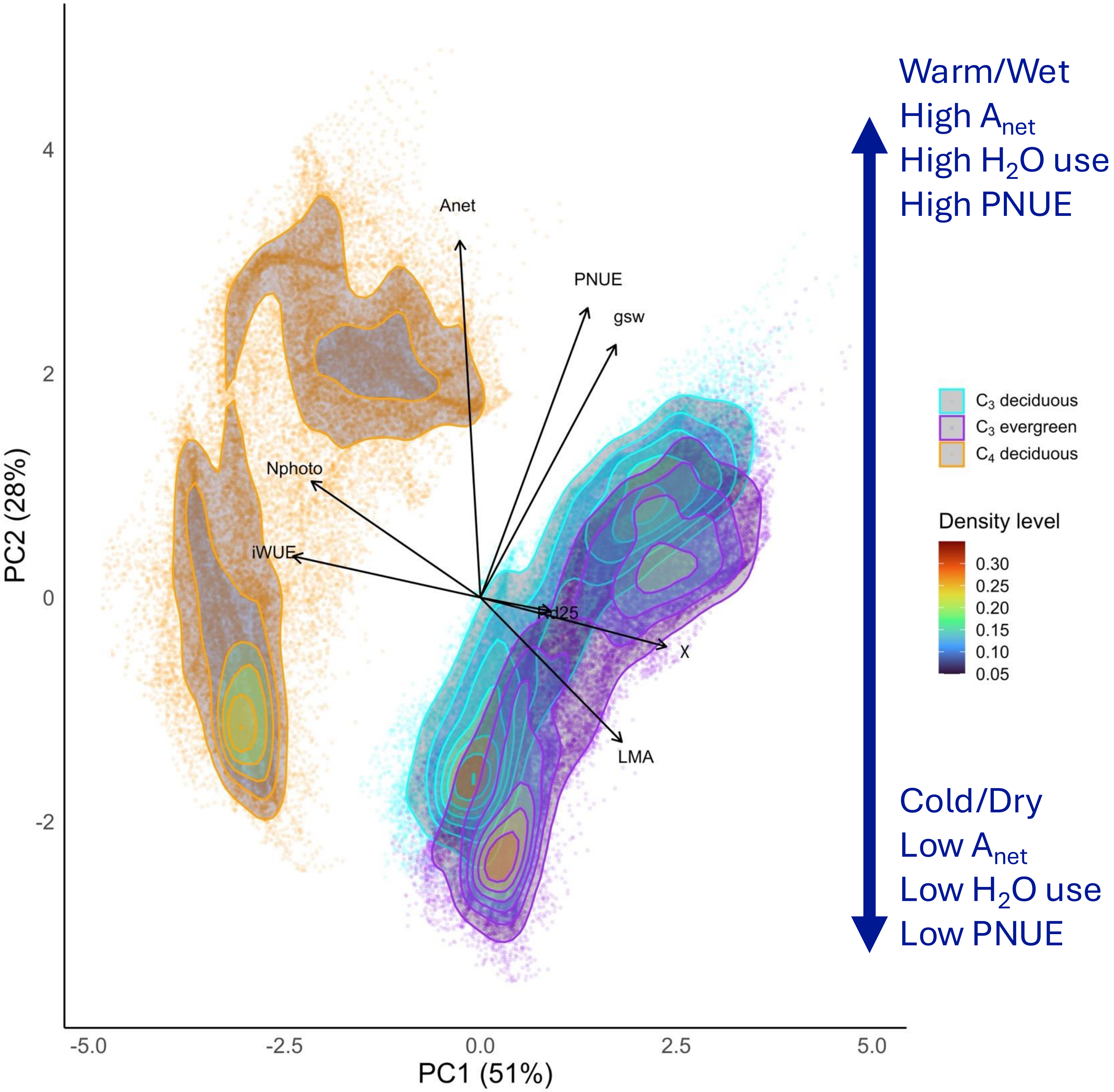


Anet
PNUE
gsw
Nphoto
iWUE
Rd25
X
LMA
PC2 (28%)
PC1 (51%)
4
2
0
-2
-5.0
-2.5
0.0
2.5
5.0
C_3 deciduous
C_3 evergreen
C_4 deciduous
Density level
0.30
0.25
0.20
0.15
0.10
0.05



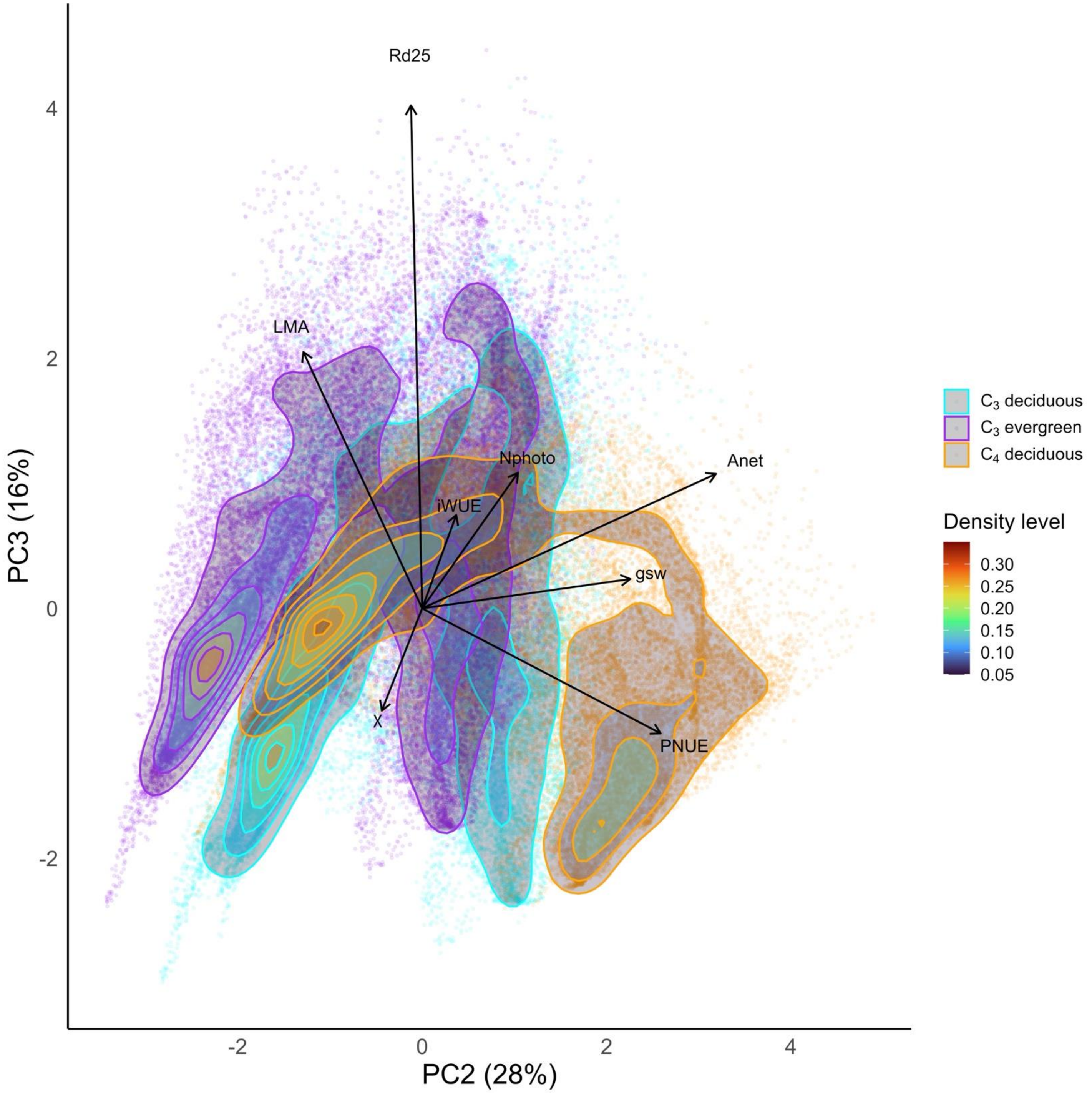
PC1 (51%)

- N use-WUE
- C_3-C_4 divide

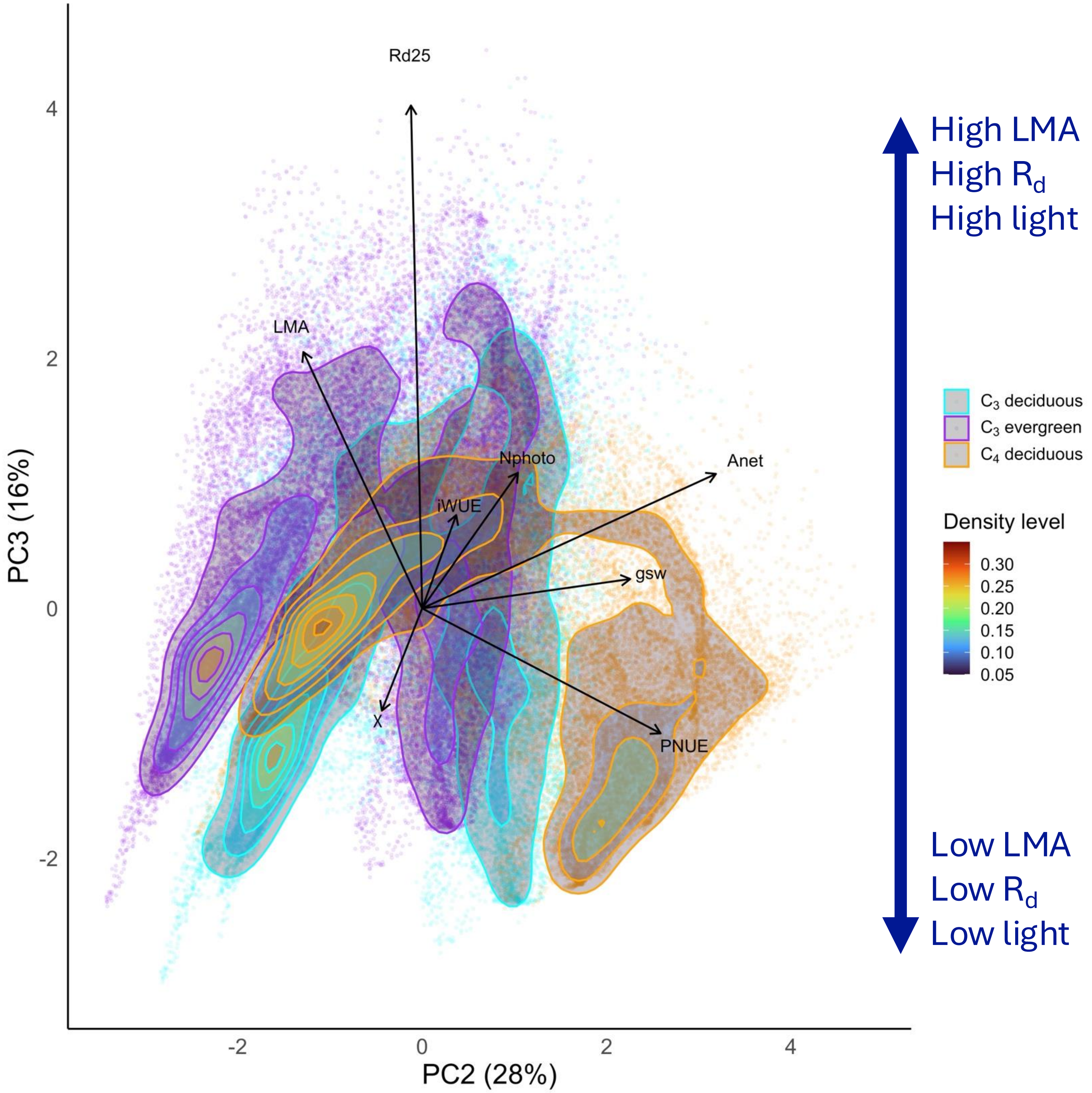


PC2 (28%)

- H_2O use-PNUE
- A_{net}
- Warm/wet-Cold/dry divide



Rd25
LMA
Nphoto
Anet
iWUE
gsw
X
PNUE
PC3 (16%)
PC2 (28%)
4
2
0
-2
-2
0
2
4
C3 deciduous
C3 evergreen
C4 deciduous
Density level
0.30
0.25
0.20
0.15
0.10
0.05



PC3 (16%)

- LMA, R_d (C costs)
- Light intensity

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Site-level simulation approach

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- Chose sites representing different dominant PFTs

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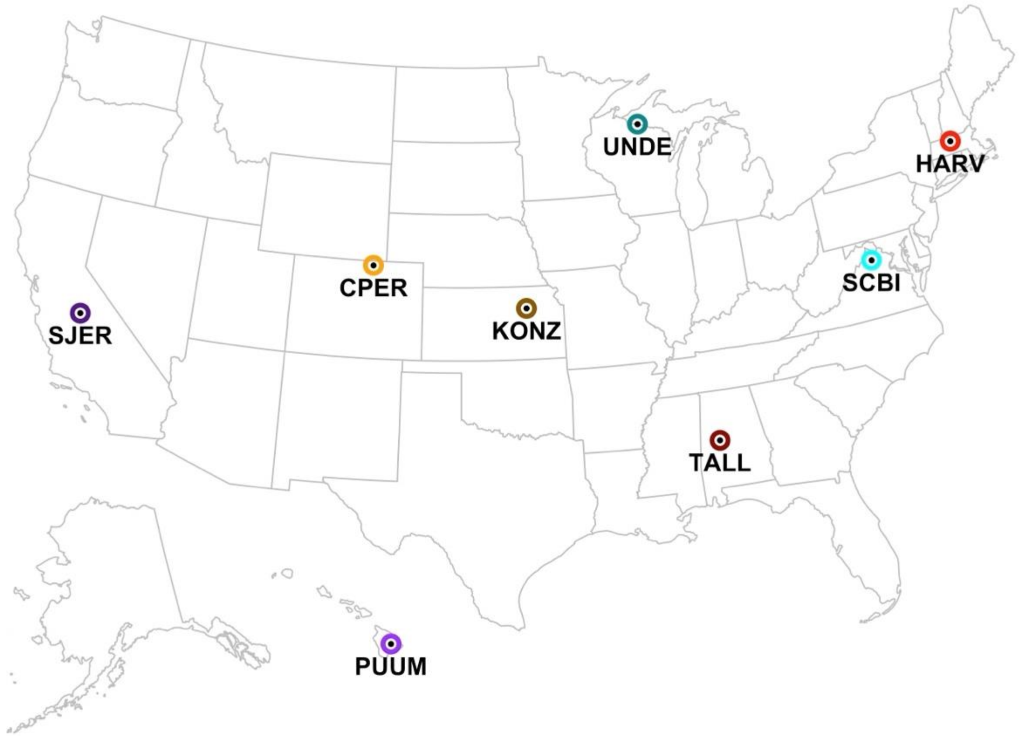
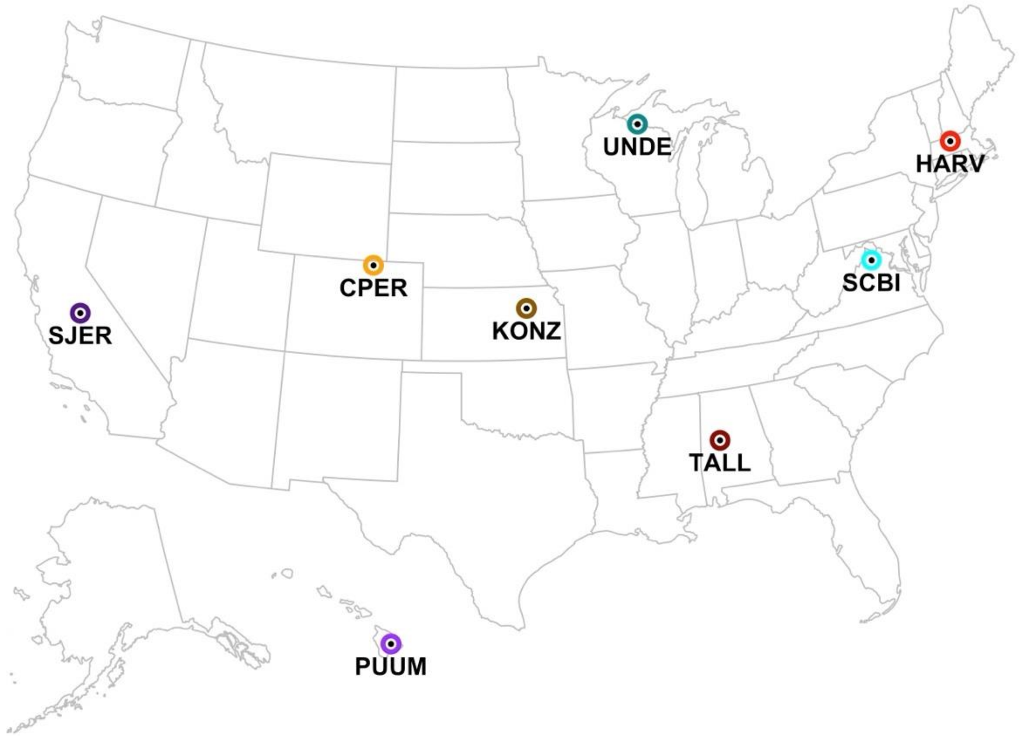
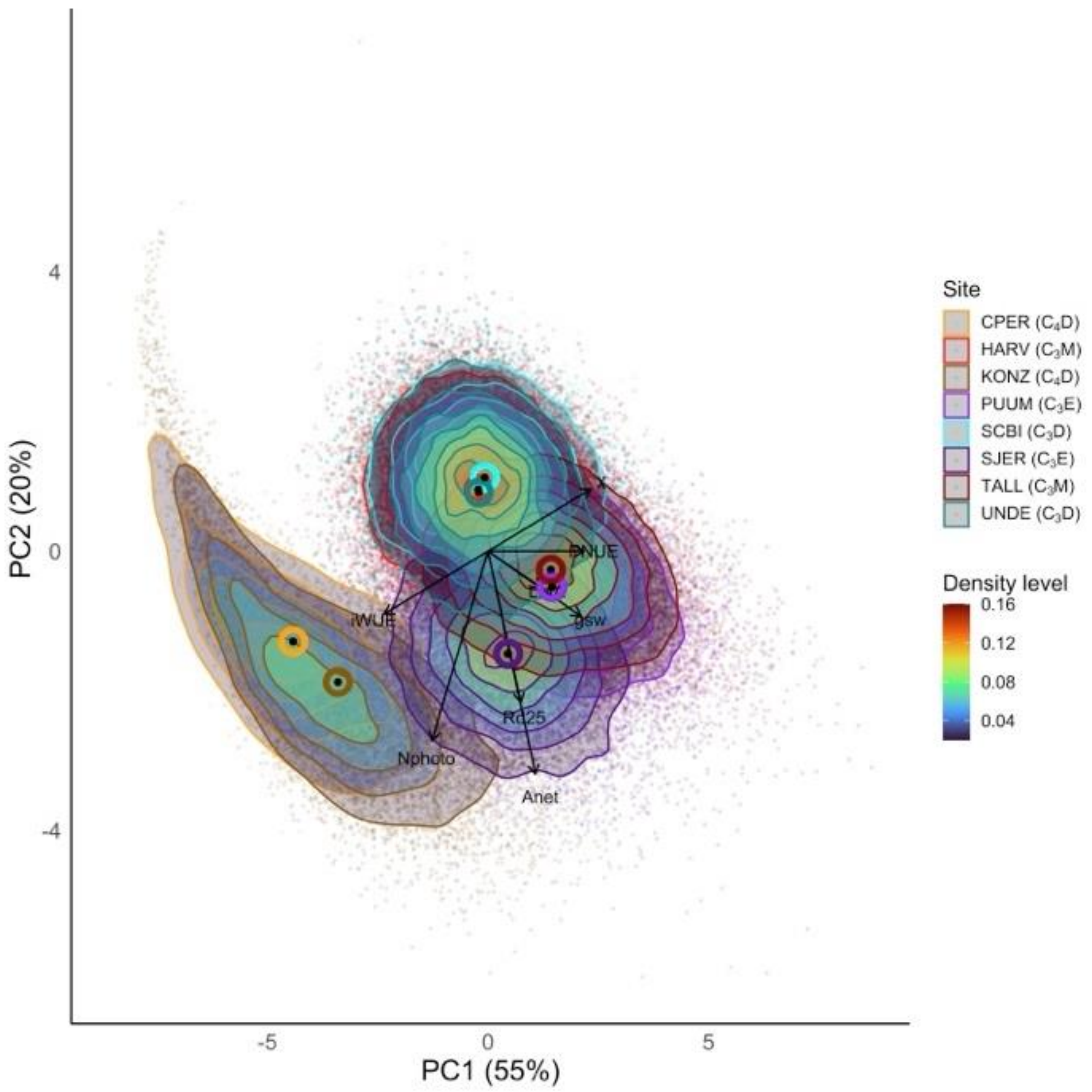
  - NEON network
    1. Deciduous forests
    2. Evergreen forests
    3. Mixed forests
    4. C_4 grasslands



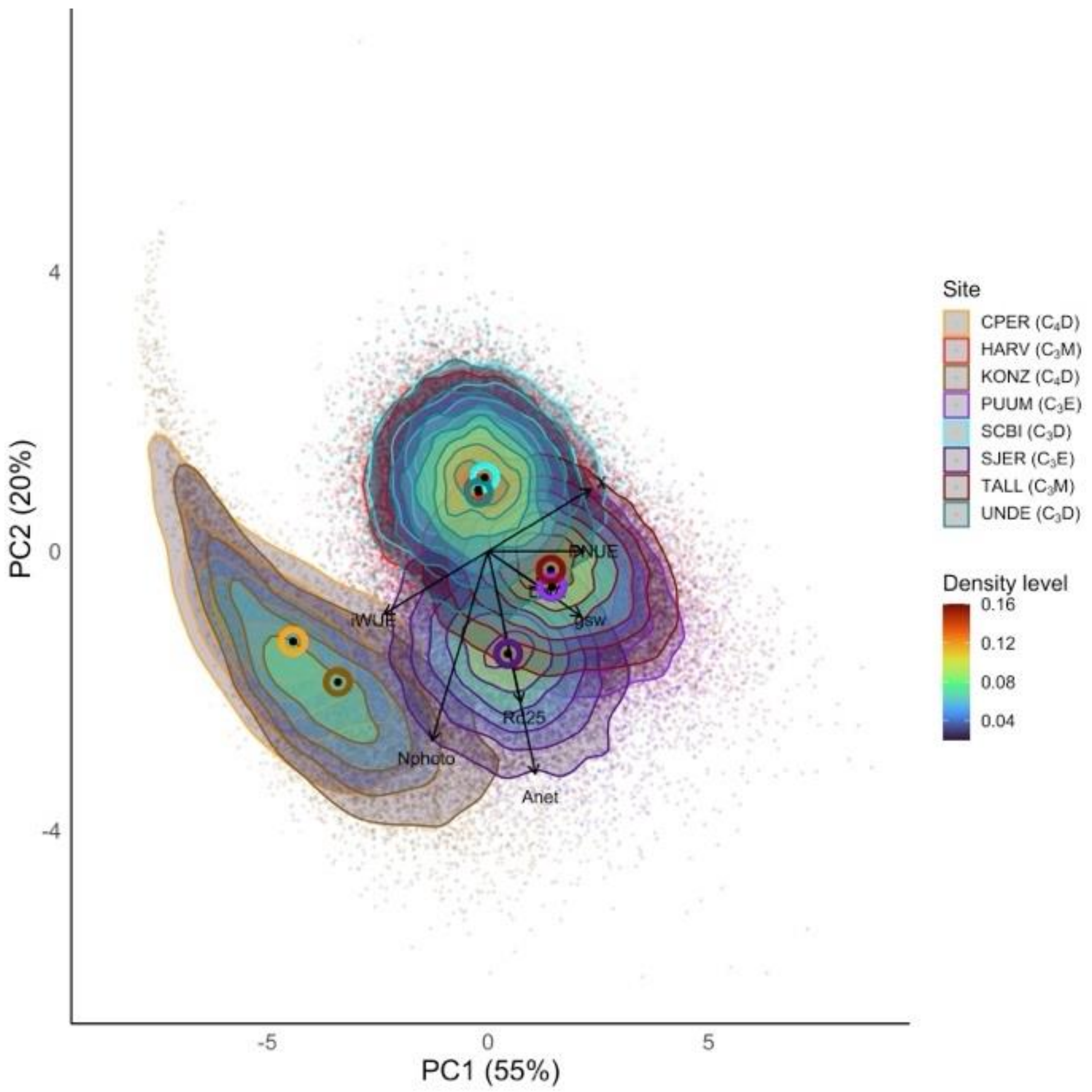
Site-level simulation approach

- Chose sites representing different dominant PFTs
  - NEON network
    1. Deciduous forests
    2. Evergreen forests
    3. Mixed forests
    4. C_4 grasslands
- Simulated optimal traits 10,000x using climate distribution
  - mean +/- 0.1*mean

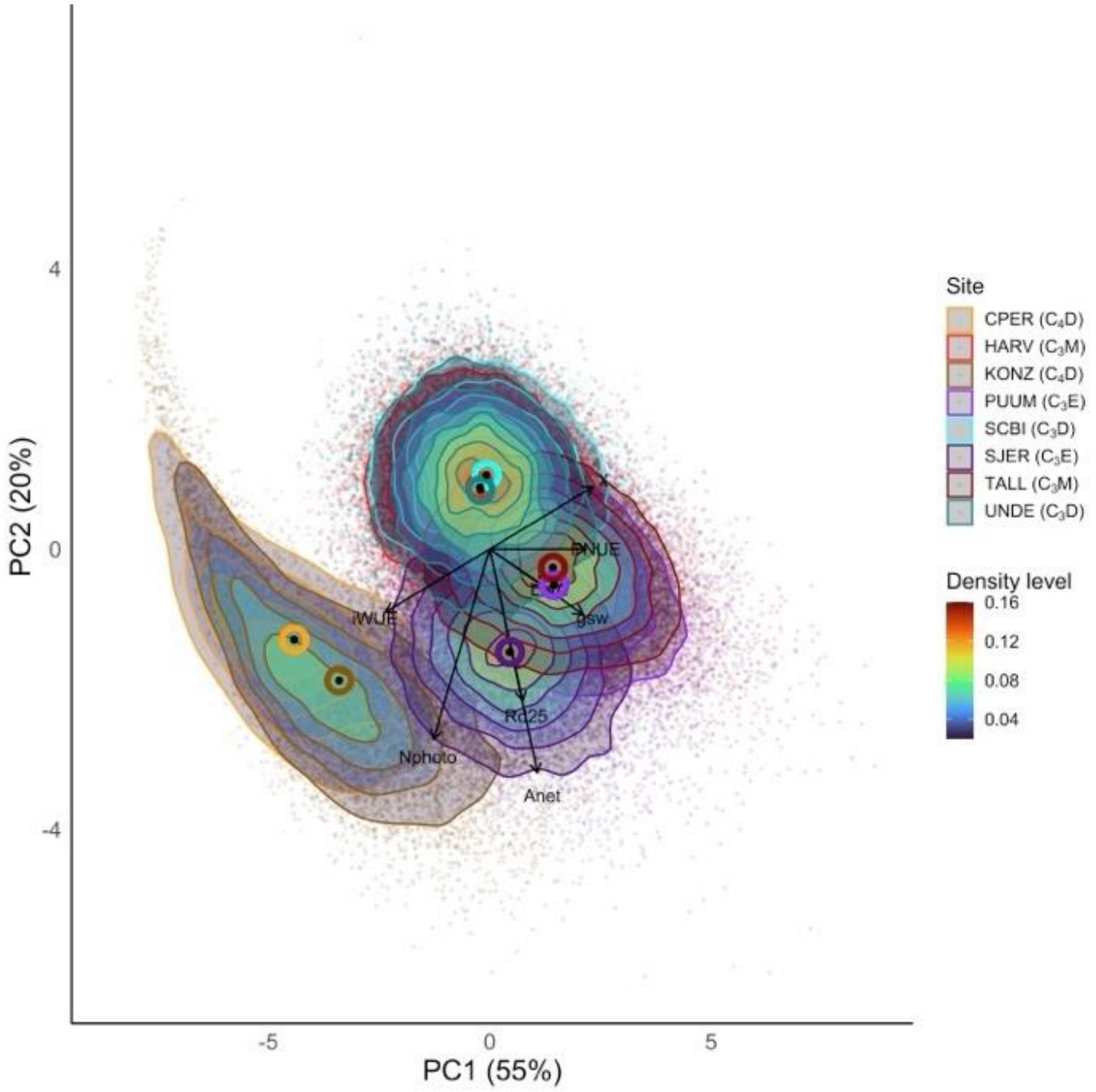




PC2 (20%)
PC1 (55%)
4
0
-4
-5
0
5
PNUE
iWUE
gsw
Rd25
Nphoto
Anet
Site
CPER (C_4D)
HARV (C_3M)
KONZ (C_4D)
PUUM (C_3E)
SCBI (C_3D)
SJER (C_3E)
TALL (C_3M)
UNDE (C_3D)
Density level
0.16
0.12
0.08
0.04

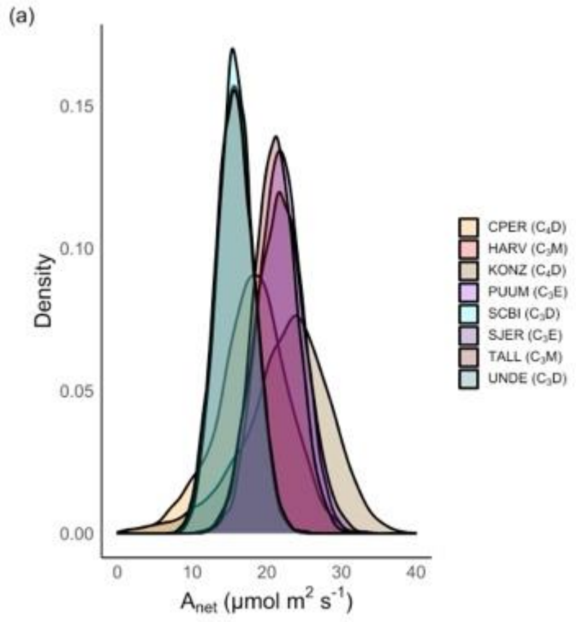


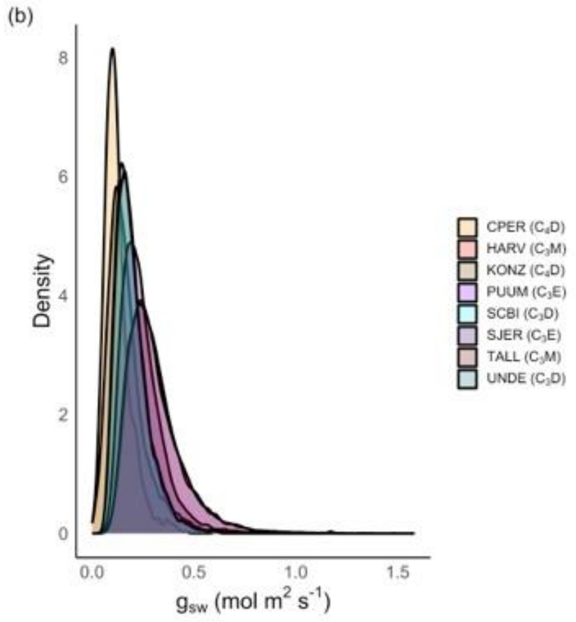
PCs defined similarly to global simulation

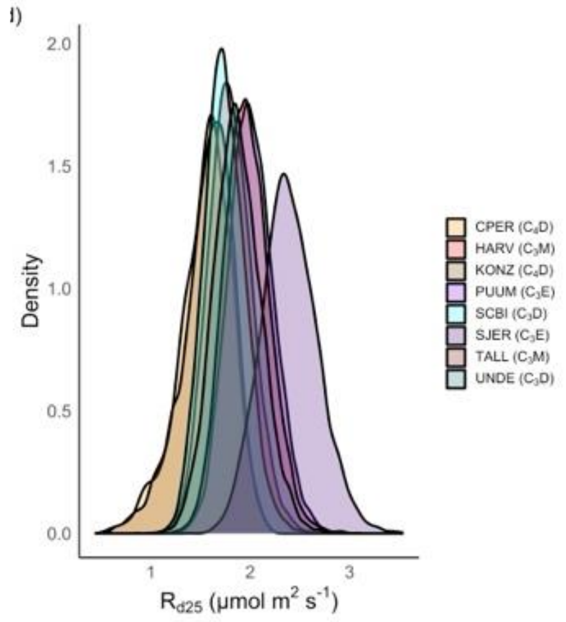


PCs defined similarly to global simulation

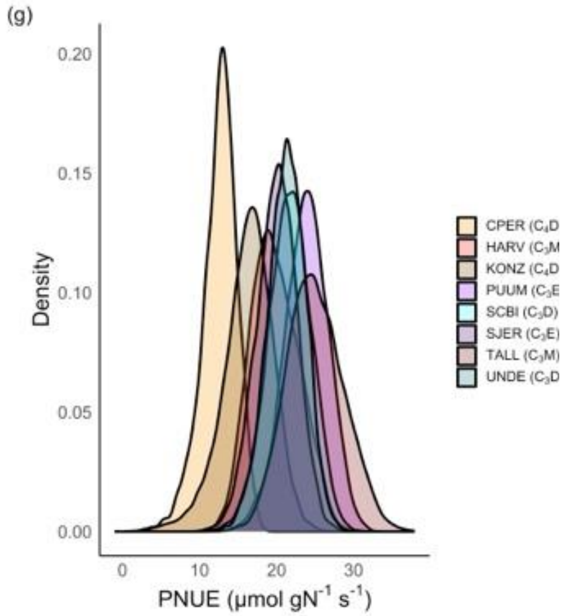
A lot of within-site variability!



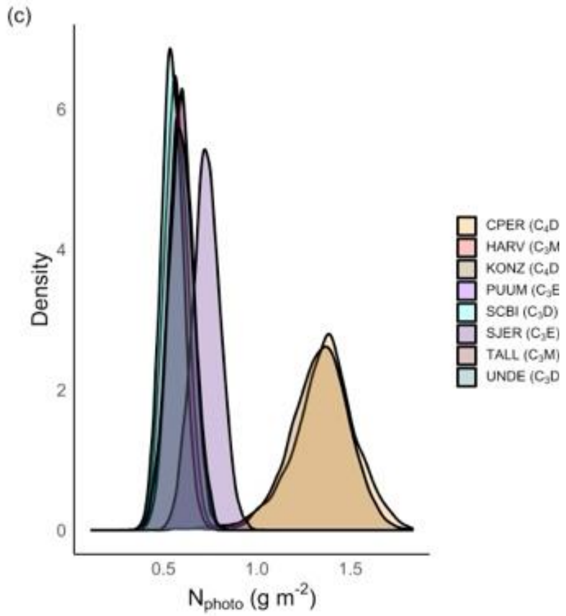


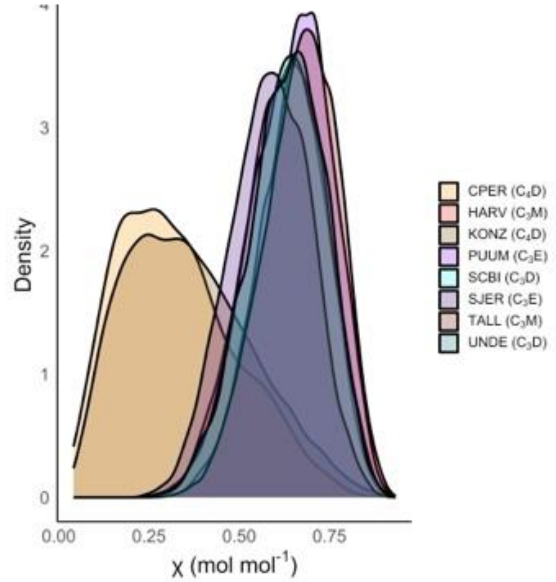


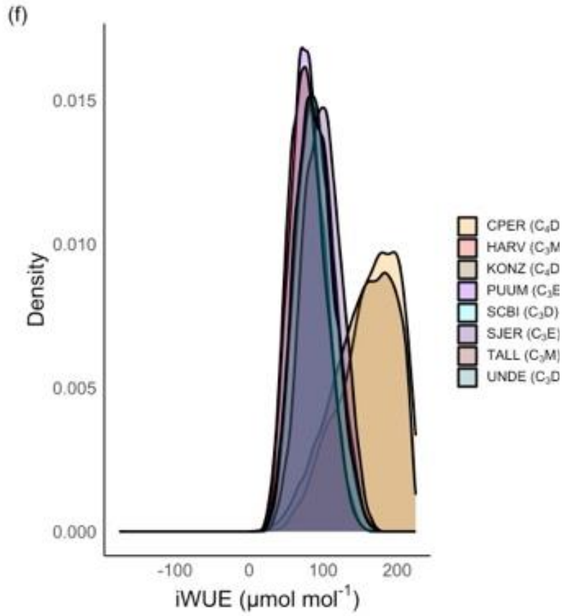
Within-site variability is larger than across-site for many traits



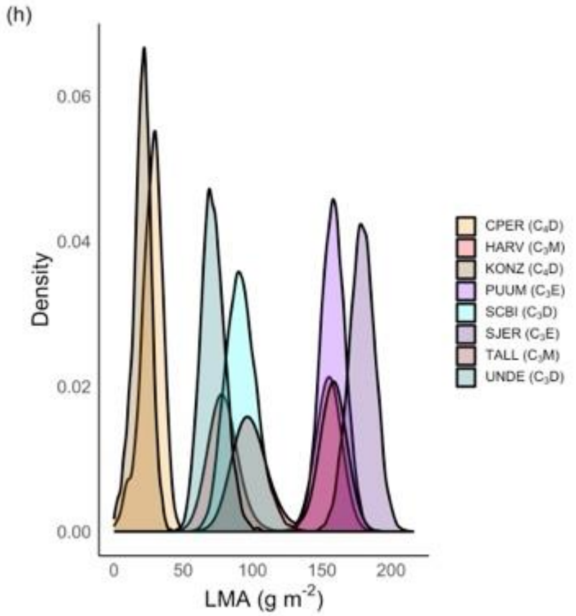
Some traits show separation by photosynthetic type







LMA shows separation by PFT



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    1. Nitrogen use-water use efficiency (51%; PFT-driven)

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    2. Water use-nitrogen use efficiency (28%; climate-driven)

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    1. Nitrogen use-water use efficiency (51%; PFT-driven)
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    3. Tissue costs of photosynthesis (16%; light-driven)

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    1. Nitrogen use-water use efficiency (51%; PFT-driven)
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- Plant type dictates optimal photosynthetic strategies via some (LMA, WUE, N use), but not all traits

Implications

- We can simulate optimal photosynthetic strategies
- Three defining axes (95%) of optimal photosynthetic strategies:
    1. Nitrogen use-water use efficiency (51%; PFT-driven)
    2. Water use-nitrogen use efficiency (28%; climate-driven)
    3. Tissue costs of photosynthesis (16%; light-driven)
- Plant type dictates optimal photosynthetic strategies via some (LMA, WUE, N use), but not all traits
- There are many optimal strategies within an environment

There are many ways plants can optimize photosynthesis and we can now quantitatively predict those strategies

Quick notice

- Lab is moving to the University of Nebraska - Lincoln
- Starting Fall 2026
- Looking for lab manager and graduate students
- Reach out if you are interested in collaborating

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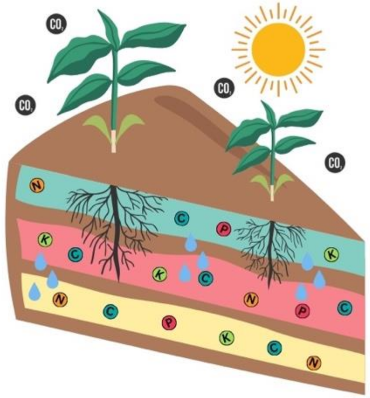
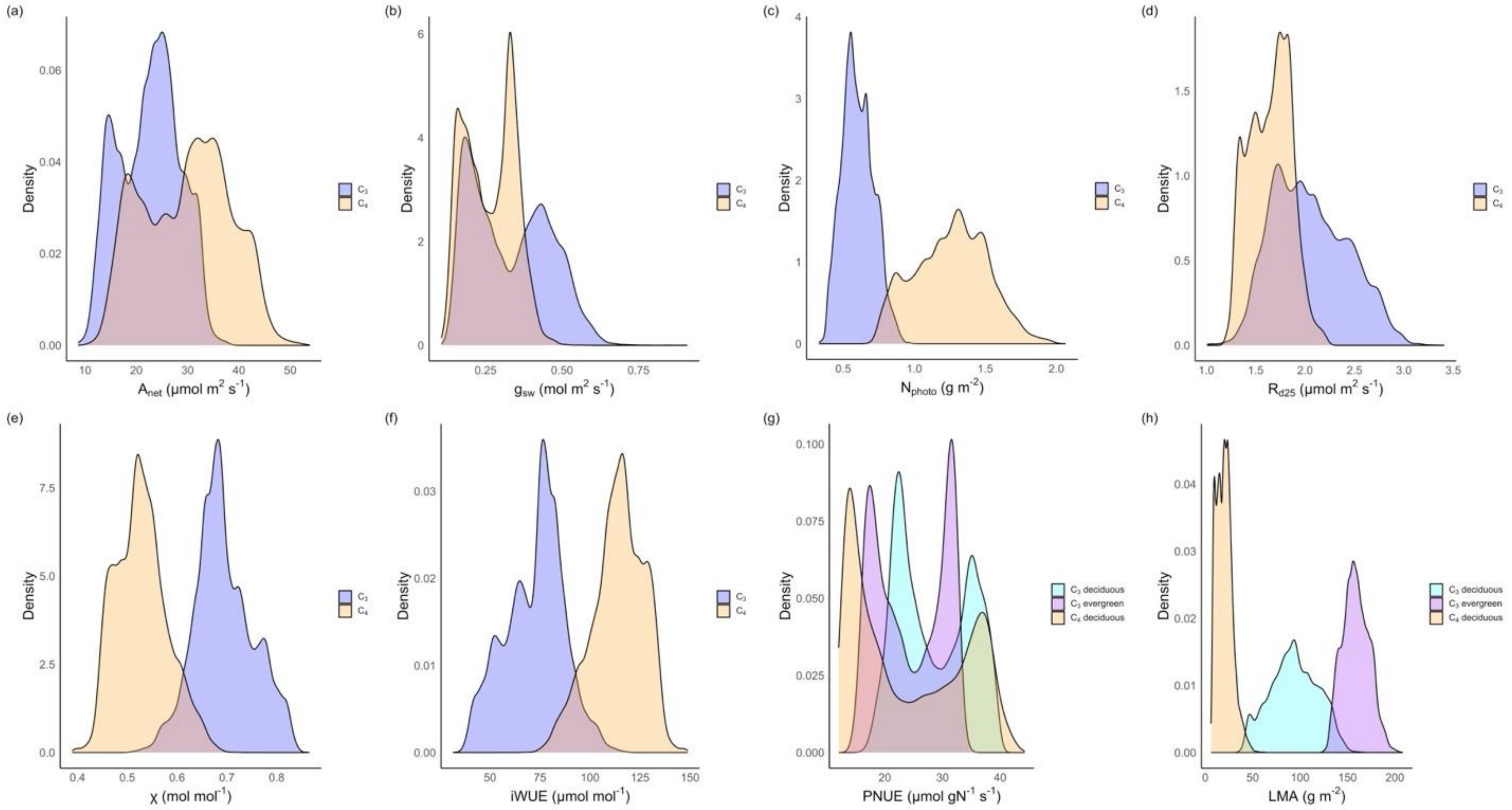
LINCOLN



Presentation: github.com/SmithEcophysLab/esa_2026
Model code: github.com/SmithEcophysLab/optimal_vcmax_r
Contact: nick.smith@ttu.edu; nsmith87@unl.edu

Schmidt Sciences



(a)
Density
0.00
0.02
0.04
0.06
10 20 30 40 50
A_{net} (μmol m^2 s^{-1})
C$_3$
C$_4$
(b)
Density
0 2 4 6
0.25 0.50 0.75
g_{sw} (mol m^2 s^{-1})
C$_3$
C$_4$
(c)
Density
0 1 2 3 4
0.5 1.0 1.5 2.0
N_{photo} (g m^{-2})
C$_3$
C$_4$
(d)
Density
0.0 0.5 1.0 1.5
1.0 1.5 2.0 2.5 3.0 3.5
R_{d25} (μmol m^2 s^{-1})
C$_3$
C$_4$
(e)
Density
0.0 2.5 5.0 7.5
0.4 0.5 0.6 0.7 0.8
χ (mol mol^{-1})
C$_3$
C$_4$
(f)
Density
0.00 0.01 0.02 0.03
50 75 100 125 150
iWUE (μmol mol^{-1})
C$_3$
C$_4$
(g)
Density
0.000 0.025 0.050 0.075 0.100
20 30 40
PNUE (μmol gN^{-1} s^{-1})
C$_3$ deciduous
C$_3$ evergreen
C$_4$ deciduous
(h)
Density
0.00 0.01 0.02 0.03 0.04
0 50 100 150 200
LMA (g m^{-2})
C$_3$ deciduous
C$_3$ evergreen
C$_4$ deciduous